Embedding Space Visualisation

This notebook investigates how embeddings extracted from the model are organised within the embedding space.

The embedding space is analysed using several dimensionality reduction techniques, including:

- PCA
- t-SNE
- UMAP

Analysis is performed at both the image level, using CLS embeddings, and the local level, using patch embeddings. This allows investigation of how entire images relate to one another within the embedding space, as well as how individual patches relate to other patches within the same image.

Scoring metrics will be used to analyse cluster quality and separation, including:

- Silhouette Score
- Intra-Cluster Distance
- Inter-Cluster Distance

The objective of this notebook is to understand the structure of the embedding space and identify characteristics that can help with anomaly detection. These findings will be used to explain the performance of the baseline model and provide a reference for future experiments involving fine-tuning and alternative embedding representations.

CLS Embedding Projections

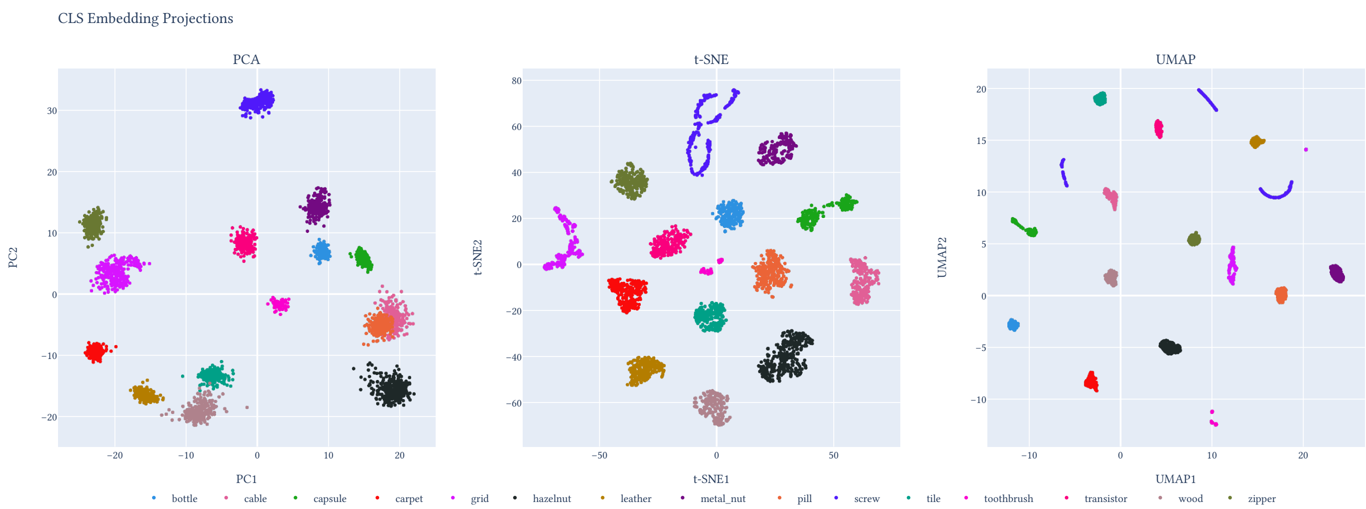
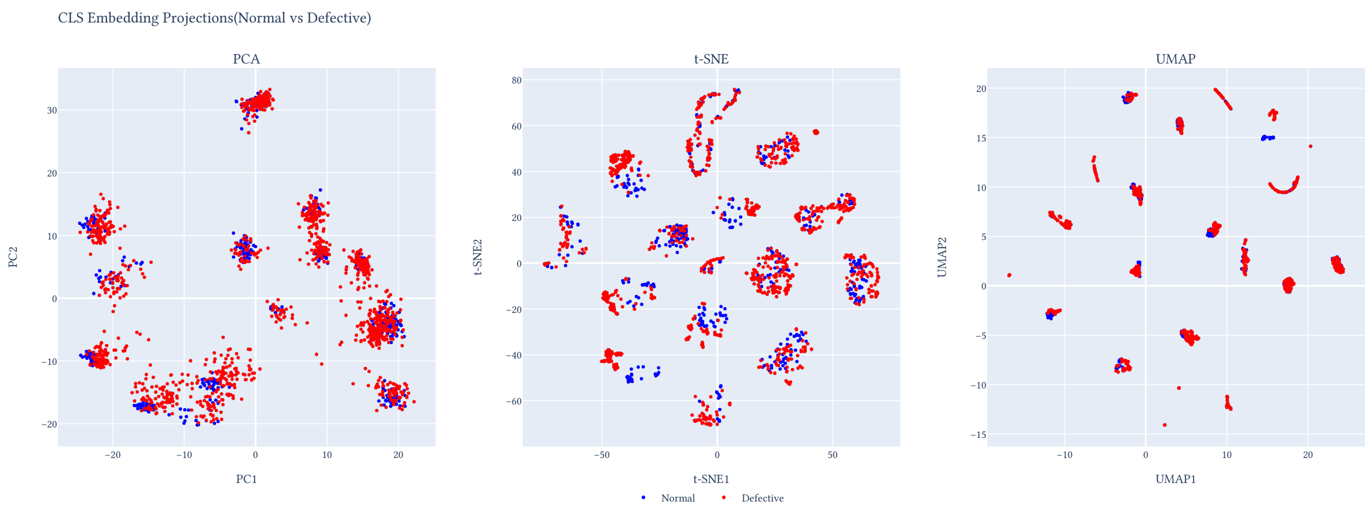


Figure 1: Embedding Projections of CLS tokens

Within the PCA projection, we can observe how the embedding space is organised along the two principal components that capture the greatest variance in the data. Several categories form distinct clusters that reflect similarities in their visual appearance. For example, relatively flat and homogeneous objects such as tile, leather, and wood are positioned closer together, while categories such as screw occupy a more distant region of the embedding space due to their more complex geometric structure and visual texture.

In the t-SNE projection, we can observe how compact or diffuse each category is within the embedding space. Most categories form relatively tight clusters, indicating that samples from the same category are represented consistently by the model. Notable exceptions include grid and screw, which exhibit more diffuse and fragmented structures. The screw category is particularly interesting, as it also achieved some of the lowest anomaly detection performance in the previous analysis. Rather than forming a single compact cluster, the embeddings are distributed in a figure-eight like structure, suggesting that normal and defective samples may occupy multiple overlapping regions of the embedding space. This reduced compactness may contribute to the lower AUROC scores observed for this category, as anomalous samples become more difficult to distinguish from normal samples using anomaly scoring methods.

Finally, the UMAP projection highlights how distinct the local neighbourhoods are within the embedding space and how compact these neighbourhoods remain. The observations regarding cluster compactness are similar to those seen in the t-SNE projection, with grid and screw again emerging as the most diffuse categories. Rather than forming a single coherent cluster, samples from these categories occupy multiple regions of the embedding space, with the screw category appearing to split into three distinct subregions. This provides further insight into the lower anomaly detection performance observed for screw in the previous analysis. Since normal samples are distributed across several disconnected regions rather than a single compact cluster, anomaly scoring methods may struggle to distinguish normal and defective samples consistently, resulting in reduced AUROC performance.



For the PCA projection, there is unsurprisingly little separation between normal and defective samples. Since PCA preserves the directions of greatest global variance, the principal components are dominated by category-level differences rather than the subtle differences introduced by anomalies. As a result, normal and defective samples largely occupy the same regions of the embedding space.

Within the t-SNE projection, we can observe how defective samples are positioned relative to local neighbourhoods of normal samples. For many categories, particularly those containing flat and visually consistent objects, normal and defective samples form relatively distinct regions. However, the screw category remains an exception. Here, normal and defective samples occupy almost identical regions of the embedding space, with little evidence of a distinct anomaly cluster. This provides a possible explanation for the lower anomaly detection performance observed previously, as the embedding representation does not naturally separate normal and defective screw samples.

The UMAP projection exhibits similar behaviour to t-SNE. For many categories, defective samples tend to occupy regions adjacent to, but partially separated from, normal samples. However, in more challenging categories such as screw, normal and defective samples continue to share the same embedding regions. This suggests that the model struggles to learn discriminative representations for these anomalies, making them difficult to separate using distance-based anomaly scoring methods.

CLS Embedding Statistics

	mean	median	std	min	max	worst_category	best_category
silhouette	0.0401	0.0425	0.0588	-0.0722	0.1615	capsule	leather
normal_inter	13.5318	12.1777	4.0829	7.3152	21.3277	wood	bottle
defect_inter	20.5365	19.4394	5.0035	13.0549	30.4626	capsule	tile
separation	1.5956	1.3713	0.45	0.9871	2.4991	screw	bottle
normal_intra	12.6352	11.4615	3.9479	6.9367	18.9459	wood	bottle
defect_intra	17.7923	16.6142	3.7904	12.5704	25.5036	capsule	tile
davies_bouldin	3.2804	2.8931	1.4754	1.5058	6.6545	screw	leather
cal_har	8.8111	6.6562	6.7326	2.6555	27.5506	screw	leather

Silhouette Score has a very low mean, indicating that when all categories are considered simultaneously, the CLS embeddings do not form highly compact and well-separated clusters. This is expected given that many categories contain visually similar objects and that silhouette scores become more difficult to interpret when a large number of classes are present.

Defect Inter-Cluster Distance is consistently larger than Good Inter-Cluster Distance, suggesting that defective samples tend to lie further from the category centroid than normal samples. This supports the assumption that anomalies occupy more distant regions of the embedding space.

The mean Separation Ratio of approximately 1.60 indicates that defective samples are, on average, around 60% further from the category centroid than normal samples. This suggests that the pretrained CLS embeddings already contain a useful anomaly signal despite no task-specific training.

Good Intra-Cluster Distance is generally lower than Defect Intra-Cluster Distance, indicating that normal samples form more compact clusters while defective samples exhibit greater variability. This is consistent with anomalies introducing additional visual variation within a category.

Several metrics identify bottle as the best-performing category, while screw, capsule, frequently appear as the most challenging categories. This aligns with the earlier AUROC analysis, where screw consistently demonstrated poorer anomaly detection performance.

Patch Embedding Statistics

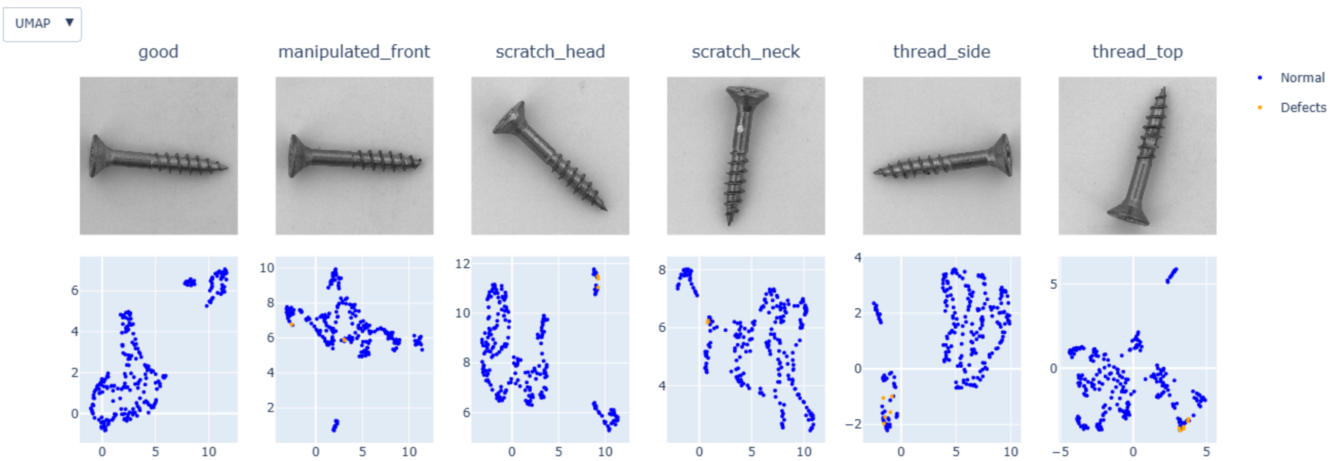
	silhouette	davies_bouldi	cal_har	normal_inter	defect_inter	separation	normal_intra
bottle	0.1032	2.0161	21.0375	35.567	38.7165	1.0888	33.8505
cable	0.1234	2.0764	17.4583	36.312	42.2481	1.1644	34.2263
capsule	0.1103	1.4781	8.0617	37.6489	45.1315	1.1989	37.0514
carpet	0.2666	1.1308	25.0733	28.0213	44.2434	1.5799	24.8757
grid	0.2727	1.1495	22.83	29.8463	44.9653	1.5073	24.6649
hazelnut	0.1479	1.4992	19.9355	39.1841	47.6354	1.2157	37.6479
leather	0.2748	0.946	14.5154	30.8956	49.6409	1.6115	26.4941
metal_nut	0.0924	2.0189	31.2216	37.5273	37.4852	0.9997	34.3597
pill	0.0977	1.7748	14.5052	37.9695	42.7971	1.1274	36.554
screw	0.1252	1.2913	5.5748	37.5862	45.9468	1.2225	35.9715
tile	0.2694	1.4006	56.9894	32.41	45.3849	1.3964	25.9535
toothbrush	0.106	1.7726	13.9147	37.6068	43.5274	1.1573	36.3193
transistor	0.086	2.7955	13.6462	32.5668	37.1644	1.1422	30.3868
wood	0.2768	1.4002	40.1426	30.474	42.6916	1.4054	23.5885
zipper	0.083	1.7521	12.1111	33.2571	37.3918	1.1236	31.261

The silhouette scores are generally much higher than those seen for the CLS embeddings, with most categories falling between 0.1 and 0.3. This suggests that patch embeddings form more coherent local structures than global image-level embeddings.

Most categories achieve separation ratios above 1, suggesting that defects are generally positioned further from the normal embedding distribution. Leather (1.61) demonstrates the strongest separation, while metal_nut (≈1.00) exhibits almost no difference between normal and defective patch distances. This indicates that defective metal nut patches occupy nearly the same embedding regions as normal patches.

Most categories have an intra-cluster distance value close to 1, suggesting that defective patches are similarly dispersed to normal patches. However, categories such as leather (1.08) show defective patches becoming more diffuse, while wood (0.82) exhibits more compact defective clusters. This suggests that anomalies affect categories differently, with some defects introducing greater variation and others producing consistent visual patterns.

Patch Embedding Projections



The screw category performs noticeably better at the patch level than at the CLS embedding level. With stats being:

- Silhoutte ≈ 0.125
- Separation ≈ 1.2
- Intra-ratio ≈ 0.97

One possible explanation is that the anomalous regions within screw images typically occupy only a small portion of the image. As a result, the global CLS embedding is dominated by information from the surrounding normal screw structure, causing normal and defective images to appear highly similar in the image-level embedding space. In contrast, patch embeddings focus on local image regions, allowing patches containing anomalies to be represented separately from normal patches. This suggests that local representations are better able to capture the subtle visual differences associated with screw defects.